Quality Report

Generated with Pix4Ddiscovery version 4.8.4

Important: Click on the different icons for:

Help to analyze the results in the Quality Report

Additional information about the sections

Click here for additional tips to analyze the Quality Report

Summary

Project	m5
Processed	2023-08-13 11:18:25
Camera Model Name(s)	FLIR_7.5_640x512 (Thermal IR)
Average Ground Sampling Distance (GSD)	7.47 cm / 2.94 in
Area Covered	0.035 km^2 / 3.4572 ha / 0.01 sq. mi. / 8.5475 acres
Time for Initial Processing (without report)	03m:26s

Quality Check

Images	median of 5916 keypoints per image	⚠
Dataset	223 out of 226 images calibrated (98%), all images enabled, 2 blocks	⚠
Camera Optimization	1.71% relative difference between initial and optimized internal camera parameters	✔
Matching	median of 2266.78 matches per calibrated image	✔
Georeferencing	yes, no 3D GCP	⚠

Preview

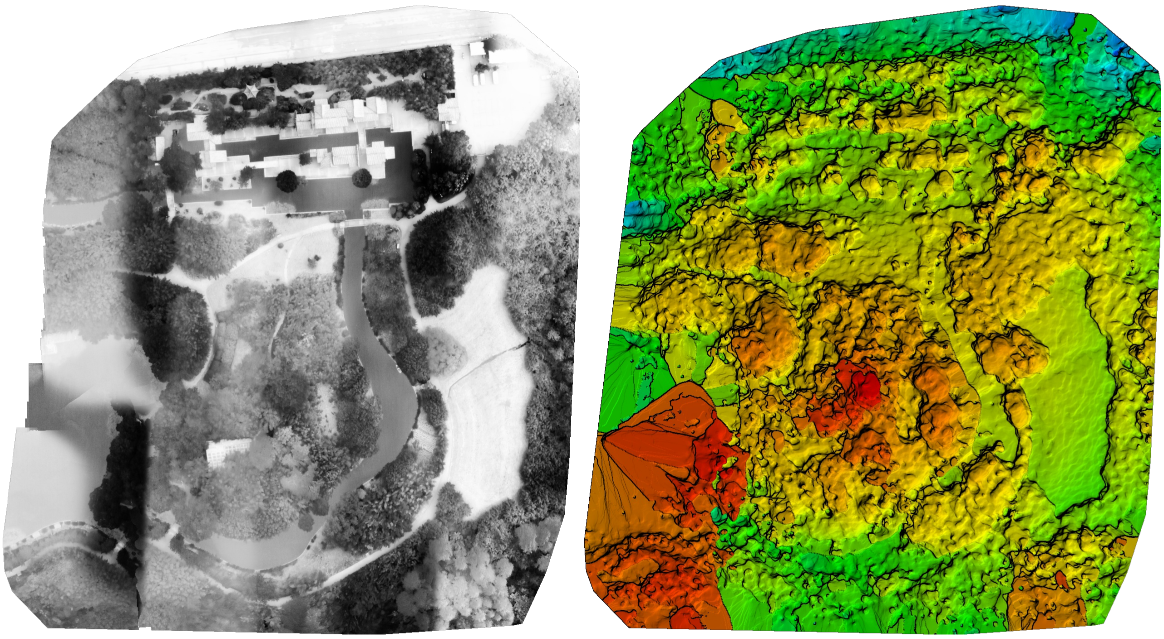


Figure 1: Orthomosaic and the corresponding sparse Digital Surface Model (DSM) before densification.

Calibration Details

Number of Calibrated Images	223 out of 226
Number of Geolocated Images	226 out of 226

Initial Image Positions

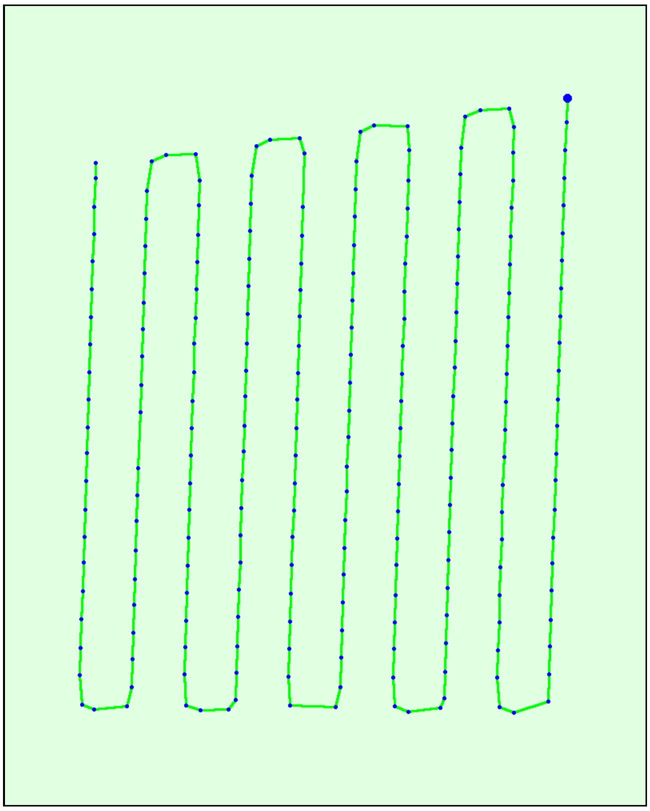
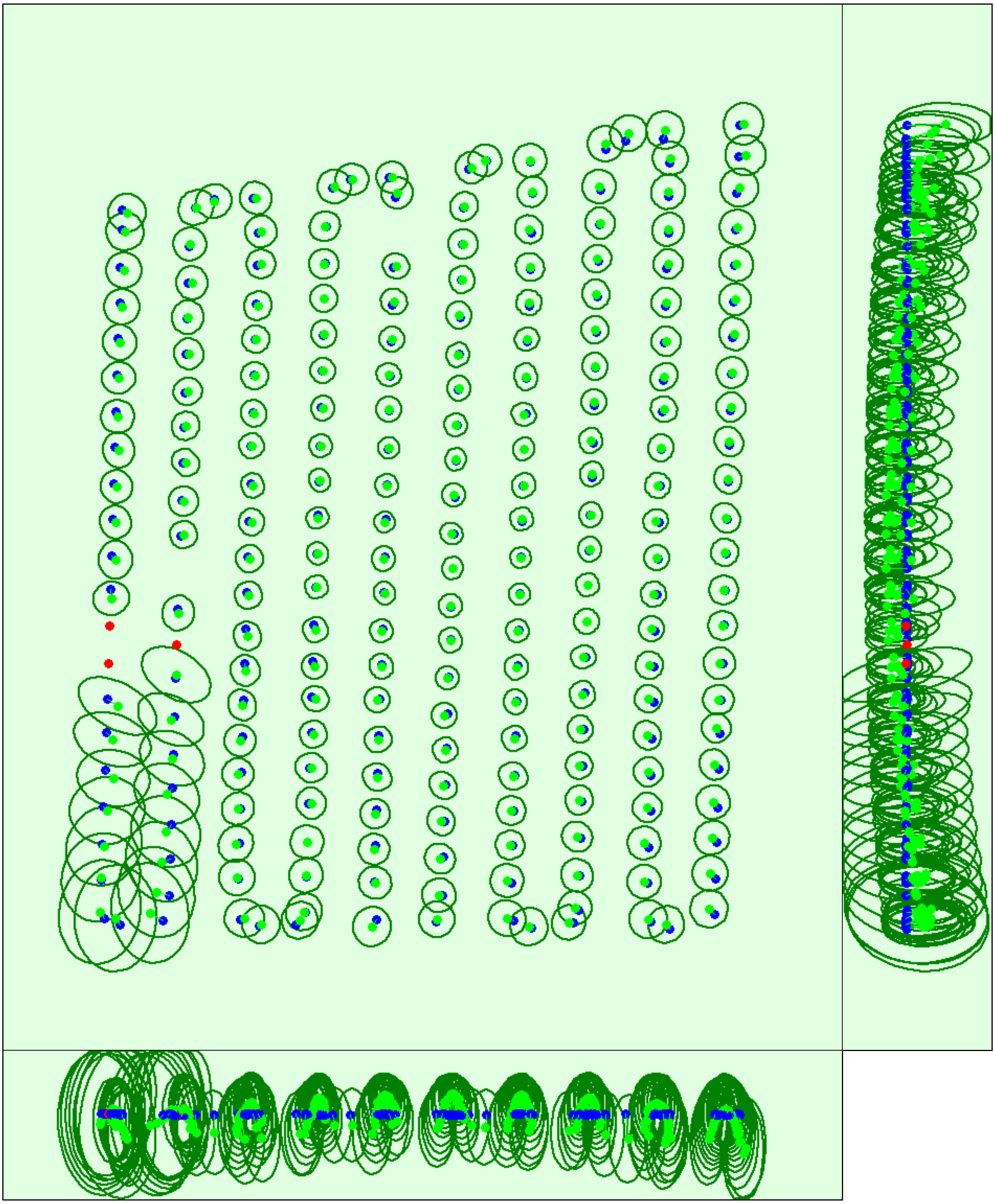


Figure 2: Top view of the initial image position. The green line follows the position of the images in time starting from the large blue dot.

Computed Image/GCPs/Manual Tie Points Positions



Uncertainty ellipses 10x magnified

Figure 3: Offset between initial (blue dots) and computed (green dots) image positions as well as the offset between the GCPs initial positions (blue crosses) and their computed positions (green crosses) in the top-view (XY plane), front-view (XZ plane), and side-view (YZ plane). Red dots indicate disabled or uncalibrated images. Dark green ellipses indicate the absolute position uncertainty of the bundle block adjustment result.

Absolute camera position and orientation uncertainties

	X[m]	Y[m]	Z[m]	Omega [degree]	Phi [degree]	Kappa [degree]
Mean	0.417	0.420	0.908	0.520	0.623	0.263
Sigma	0.158	0.168	0.296	0.059	0.049	0.161

Overlap

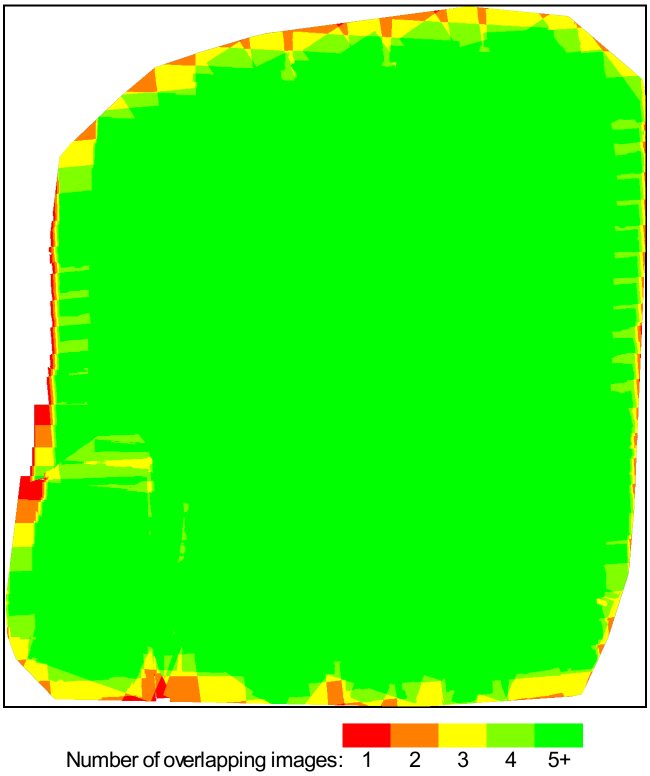


Figure 4: Number of overlapping images computed for each pixel of the orthomosaic.
Red and yellow areas indicate low overlap for which poor results may be generated. Green areas indicate an overlap of over 5 images for every pixel. Good quality results will be generated as long as the number of keypoint matches is also sufficient for these areas (see Figure 5 for keypoint matches).

Bundle Block Adjustment Details

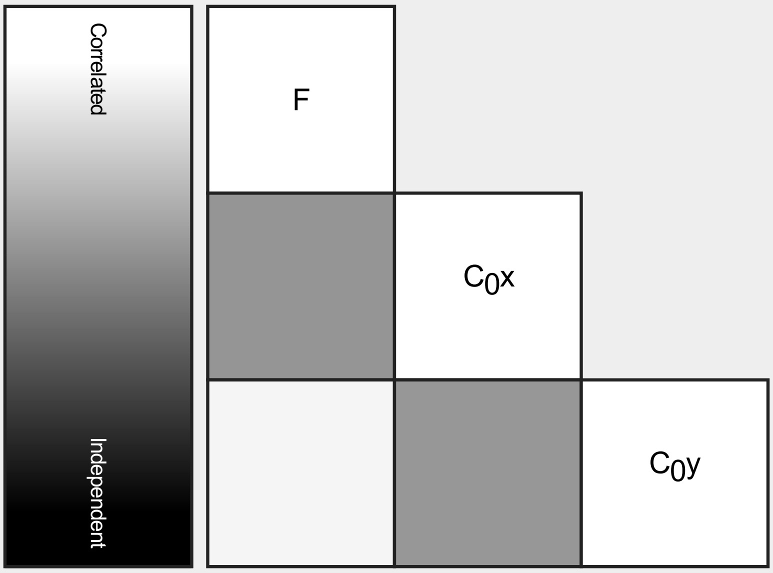
Number of 2D Keypoint Observations for Bundle Block Adjustment	569749
Number of 3D Points for Bundle Block Adjustment	190123
Mean Reprojection Error [pixels]	0.424

Internal Camera Parameters

FLIR_7.5_640x512 (Thermal IR). Sensor Dimensions: 10.880 [mm] x 8.704 [mm]

EXIF ID: _0.0_640x512

	Focal Length	Principal Point x	Principal Point y	R1	R2	R3	T1	T2
Initial Values	1117.650 [pixel] 19.000 [mm]	320.000 [pixel] 5.440 [mm]	256.000 [pixel] 4.352 [mm]	0	0	0	0	0
Optimized Values	1136.836 [pixel] 19.326 [mm]	321.153 [pixel] 5.460 [mm]	248.000 [pixel] 4.216 [mm]	0	0	0	0	0
Uncertainties (Sigma)	6.322 [pixel] 0.107 [mm]	0.354 [pixel] 0.006 [mm]	1.046 [pixel] 0.018 [mm]					



The correlation between camera internal parameters determined by the bundle adjustment. White indicates a full correlation between the parameters, ie. any change in one can be fully compensated by the other. Black indicates that the parameter is completely independent, and is not affected by other parameters.

The number of Automatic Tie Points (ATPs) per pixel, averaged over all images of the camera model, is color coded between black and white. White indicates that, on average, more than 16 ATPs have been extracted at the pixel location. Black indicates that, on average, 0 ATPs have been extracted at the pixel location. Click on the image to the see the average direction and magnitude of the re-projection error for each pixel. Note that the vectors are scaled for better visualization. The scale bar indicates the magnitude of 1 pixel error.

2D Keypoints Table

	Number of 2D Keypoints per Image	Number of Matched 2D Keypoints per Image
Median	5916	2267
Min	3595	366
Max	8975	5319
Mean	6057	2555

3D Points from 2D Keypoint Matches

	Number of 3D Points Observed
In 2 Images	115465
In 3 Images	34571
In 4 Images	16336
In 5 Images	8702
In 6 Images	4250
In 7 Images	3047
In 8 Images	2210
In 9 Images	1662
In 10 Images	1201
In 11 Images	786
In 12 Images	553
In 13 Images	435
In 14 Images	353
In 15 Images	260
In 16 Images	135
In 17 Images	64
In 18 Images	43
In 19 Images	32
In 20 Images	12
In 21 Images	6

2D Keypoint Matches

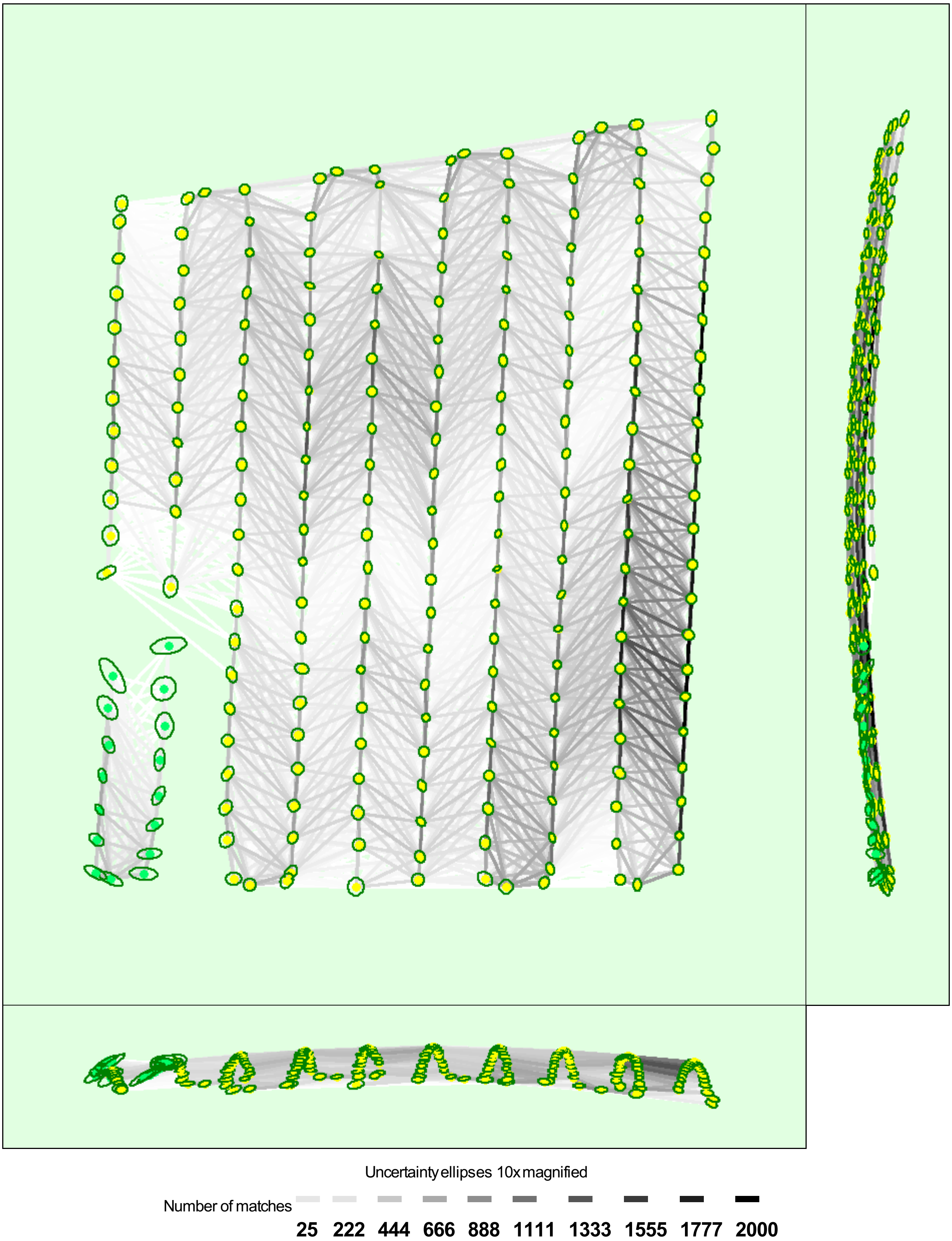


Figure 5: Computed image positions with links between matched images. The darkness of the links indicates the number of matched 2D keypoints between the images. Bright links indicate weak links and require manual tie points or more images. Dark green ellipses indicate the relative camera position uncertainty of the bundle block adjustment result.

Relative camera position and orientation uncertainties

	X [m]	Y [m]	Z [m]	Omega [degree]	Phi [degree]	Kappa [degree]
Mean	0.145	0.153	0.064	0.141	0.116	0.082

Sigma	0.047	0.044	0.036	0.101	0.066	0.174

Geolocation Details

Absolute Geolocation Variance

Min Error [m]	Max Error [m]	Geolocation Error X [%]	Geolocation Error Y [%]	Geolocation Error Z [%]
-	-15.00	0.00	0.00	0.00
-15.00	-12.00	0.00	0.00	0.00
-12.00	-9.00	0.00	0.00	0.00
-9.00	-6.00	0.00	0.00	0.00
-6.00	-3.00	0.00	0.00	20.18
-3.00	0.00	51.57	55.16	34.98
0.00	3.00	47.53	44.84	26.46
3.00	6.00	0.90	0.00	13.90
6.00	9.00	0.00	0.00	4.04
9.00	12.00	0.00	0.00	0.45
12.00	15.00	0.00	0.00	0.00
15.00	-	0.00	0.00	0.00
Mean [m]		0.014766	-0.012298	-0.019367
Sigma [m]		0.729320	0.746024	3.172715
RMS Error [m]		0.729469	0.746125	3.172774

Min Error and Max Error represent geolocation error intervals between -1.5 and 1.5 times the maximum accuracy of all the images. Columns X, Y, Z show the percentage of images with geolocation errors within the predefined error intervals. The geolocation error is the difference between the initial and computed image positions. Note that the image geolocation errors do not correspond to the accuracy of the observed 3D points.

Relative Geolocation Variance

Relative Geolocation Error	Images X [%]	Images Y [%]	Images Z [%]
[-1.00, 1.00]	100.00	100.00	100.00
[-2.00, 2.00]	100.00	100.00	100.00
[-3.00, 3.00]	100.00	100.00	100.00
Mean of Geolocation Accuracy [m]	5.000000	5.000000	10.000000
Sigma of Geolocation Accuracy [m]	0.000000	0.000000	0.000000

Images X, Y, Z represent the percentage of images with a relative geolocation error in X, Y, Z.

Initial Processing Details

System Information

Hardware	CPU: 13th Gen Intel(R) Core(TM) i5-13500H RAM: 16GB GPU: Intel(R) Iris(R) Xe Graphics (Driver: 31.0.101.4146), Virtual Display Device (Driver: 1.0.0.9)
Operating System	Windows 11, 64-bit

Coordinate Systems

Image Coordinate System	WGS 84 (EGM96 Geoid)
Output Coordinate System	WGS 84 / UTM zone 51N (EGM96 Geoid)

Processing Options

Detected Template	Thermal Camera
Keypoints Image Scale	Full, Image Scale: 2
Advanced: Matching Image Pairs	Aerial Grid or Corridor
Advanced: Matching Strategy	Use Geometrically Verified Matching: yes
Advanced: Keypoint Extraction	Targeted Number of Keypoints: Automatic
Advanced: Calibration	Calibration Method: Alternative Internal Parameters Optimization: All External Parameters Optimization: All Rematch: Auto, yes

Point Cloud Densification details

Processing Options

Image Scale	multiscale, 1 (Original image size, Slow)
Point Density	Optimal
Minimum Number of Matches	3
3D Textured Mesh Generation	no
LOD	Generated: no
Advanced: Image Groups	Grayscale
Advanced: Use Processing Area	yes
Advanced: Use Annotations	yes
Time for Point Cloud Densification	01m:02s
Time for Point Cloud Classification	NA
Time for 3D Textured Mesh Generation	NA

Results

Number of Generated Tiles	1
Number of 3D Densified Points	1550358
Average Density (per m^3)	19.93